Artificial Intelligence (CS F407)

Tutorial Questions

Tutorial 0 – Neural Models

1. Derivative as local sensitivity and the direction of learning

Consider the scalar loss

$$L(\theta) = (\theta - 3)^2.$$

At $\theta = 5$:

(a) Interpret $\frac{dL}{d\theta}$ as a local sensitivity.
(b) Use a first-order approximation to predict the loss at $\theta = 5.01$.
(c) If gradient descent uses learning rate $\eta = 0.1$, compute one update and check whether the loss falls.
(d) Explain why backpropagation and gradient descent are not the same algorithm.

2. Multiply along paths, add across paths

A scalar s affects a loss through two routes:

$$p = 3s, \qquad q = s^2, \qquad L = p + q.$$

At $s = 2$:

(a) Compute L.
(b) Compute $\frac{dL}{ds}$ by treating the expression as a computational graph.
(c) Verify the answer by first simplifying L as a direct function of s.
(d) Explain why the "add across paths" rule matters for shared parameters in larger networks.

3. Reverse-mode automatic differentiation on a linear neuron

For one training example, let

$$z = wx, \qquad s = z + b, \qquad r = s - y, \qquad L = r^2.$$

Use

$$w = 2, \quad x = 3, \quad b = 1, \quad y = 5.$$

(a) Perform the forward pass.
(b) Starting from $\bar{L} = \frac{\partial L}{\partial L} = 1$, propagate reverse-mode sensitivities and obtain $\frac{\partial L}{\partial w}$ and $\frac{\partial L}{\partial b}$.
(c) Explain why reverse mode is attractive when a model has many parameters but only one scalar loss.

4. Why depth needs nonlinearity: the XOR case

A network is formed by stacking affine layers,

$$\boldsymbol{h}^{(\ell)} = W^{(\ell)}\boldsymbol{h}^{(\ell-1)} + \boldsymbol{b}^{(\ell)}, \qquad \boldsymbol{h}^{(0)} = \boldsymbol{x},$$

with no nonlinear activation between layers.

(a) Show for two layers that the whole network is still one affine map.
(b) Explain why adding more such affine layers cannot solve XOR with a single linear decision boundary.
(c) Show how two nonlinear hidden units can construct useful intermediate features for XOR using a step activation.

5. Softmax, cross-entropy, and the logit gradient

A hidden layer has already produced

$$\boldsymbol{h} = \begin{bmatrix} 0.6225 \\ 0.3430 \end{bmatrix}.$$

A three-class output layer uses

$$W^{(2)} = \begin{bmatrix} 0.3 & -0.5 \\ -0.2 & 0.4 \\ 0.1 & 0.2 \end{bmatrix}, \qquad \boldsymbol{b}^{(2)} = \begin{bmatrix} 0.2 \\ 0.0 \\ -0.1 \end{bmatrix}.$$

The logits and softmax probabilities are approximately

$$\boldsymbol{z} = (0.2152, 0.0127, 0.0308), \qquad \boldsymbol{p} = (0.3776, 0.3084, 0.3140).$$

The true class is class 1, so $\boldsymbol{y} = (1, 0, 0)$.

(a) Compute the cross-entropy loss.
(b) Use the softmax–cross-entropy simplification to compute $\frac{\partial \mathcal{L}}{\partial \boldsymbol{z}}$.
(c) Interpret the sign of each component.
(d) Compute $\frac{\partial \mathcal{L}}{\partial W^{(2)}}$.
(e) Explain why adding the same constant to all three logits does not change the prediction probabilities.

Tutorial 1 – Intelligent Agents

1. Constructing an Agent Function

A simple robot vacuum has the following condition–action rules.

IF Dirty	→ Vacuum
IF Wall	→ Turn Left
OTHERWISE	→ Move Forward

Suppose the robot receives the following percept sequence

(Clean, Clean, Wall, Dirty).

What action should the agent perform now?

2. Utility-Based Decision Making

An autonomous taxi must take a passenger to the airport.
The objective is simply to reach the airport, and all three routes below will eventually do so.
However, the routes differ in their likelihood of arriving on time.

Route	Probability of On-Time Arrival	Utility if On Time	Utility if Late
Highway	0.70	100	20
City Roads	0.95	80	20
Scenic Route	1.00	60	60

Which route should a *utility-based* agent choose?
How would this differ from the decision made by a *goal-based* agent?

3. Which Agent Architecture?

For each of the following systems, identify the most appropriate agent architecture.

System	Architecture
Automatic sliding door	?
GPS navigation system	?
Robot vacuum with map	?
AlphaGo	?

4. Designing a PEAS Description

Construct a PEAS description for an autonomous taxi.

5. Designing an LLM Agent

A user asks

"Find the cheapest flight from Goa to Delhi next Tuesday."

Design a simple agent architecture capable of solving this problem.

Tutorial 2 – Bayesian Networks

1. Calculating a Joint Probability

Consider the following Bayesian network:

$$B \longrightarrow A \longleftarrow E$$

Here B denotes burglary, E denotes earthquake, and A denotes alarm.

Suppose

$$P(B = t) = 0.001, \qquad P(E = t) = 0.002.$$

The conditional probabilities for the alarm are

B	E	$P(A = t \mid B, E)$
t	t	0.95
t	f	0.94
f	t	0.29
f	f	0.001

Calculate

$$P(B = t, E = t, A = t).$$

2. Calculating a Marginal Probability

Consider the simpler Bayesian network

$$R \longrightarrow T$$

Here R denotes rain and T denotes heavy traffic.

Suppose

$$P(R = t) = 0.30$$

and

$$P(T = t \mid R = t) = 0.80, \qquad P(T = t \mid R = f) = 0.20.$$

Calculate the probability that traffic is heavy:

$$P(T = t).$$

3. Reading Conditional Independence from a Bayesian Network

Consider the network

$$A \longrightarrow B \longrightarrow C.$$

Here A denotes studying, B denotes understanding the material, and C denotes passing the examination.

What conditional independence does the network suggest about A and C given B?

In other words, is

$$P(C \mid A, B) = P(C \mid B)?$$

4. Inference with a Hidden Variable

Consider the network

$$B \longrightarrow A \longleftarrow E$$

where B is burglary, E is earthquake, and A is the alarm.

Suppose

$$P(B = t) = 0.001, \qquad P(E = t) = 0.002,$$

and

B	E	$P(A = t \mid B, E)$
t	t	0.95
t	f	0.94
f	t	0.29
f	f	0.001

Suppose we know that the alarm has sounded.

Calculate

$$P(B = t \mid A = t).$$

5. The Alarm Network with Two Pieces of Evidence

Consider the full Alarm network:

$$B \longrightarrow A \longleftarrow E, \qquad A \longrightarrow J, \qquad A \longrightarrow M.$$

Here B denotes burglary, E denotes earthquake, A denotes alarm, J denotes John calls, and M denotes Mary calls.

The prior probabilities are

$$P(B = t) = 0.001, \qquad P(E = t) = 0.002.$$

The alarm CPT is

B	E	$P(A = t \mid B, E)$
t	t	0.95
t	f	0.94
f	t	0.29
f	f	0.001

The telephone-call probabilities are

$$P(J = t \mid A = t) = 0.90, \qquad P(J = t \mid A = f) = 0.05,$$

and

$$P(M = t \mid A = t) = 0.70, \qquad P(M = t \mid A = f) = 0.01.$$

John and Mary both call.

Calculate

$$P(B = t \mid J = t, M = t).$$

Tutorial 3 – Logical Intelligence

1. Clauses, Literals, and Complementary Literals

Consider the two clauses

$$C_1 = P \vee Q \vee \neg R \qquad \text{and} \qquad C_2 = \neg Q \vee R \vee S.$$

Identify:

(a) the literals in each clause;
(b) all complementary pairs of literals;
(c) one resolvent of C_1 and C_2. State which pair you used.

2. A Single Resolution Step

Suppose an AI system has the following two clauses:

$$\text{Rain} \vee \text{Snow} \qquad \text{and} \qquad \neg\text{Snow} \vee \text{Cold}.$$

Use resolution to derive a new conclusion.

3. Proving an Entailment by Refutation

An autonomous vehicle has the following knowledge:

$$\begin{aligned} &\text{WetRoad} \rightarrow \text{Slippery},\\ &\text{Slippery} \rightarrow \text{ReduceSpeed},\\ &\text{WetRoad}. \end{aligned}$$

Use resolution to establish whether ReduceSpeed follows from the knowledge base.

4. Testing Entailment by Resolution

Consider the knowledge base

$$\begin{aligned} &P \rightarrow Q,\\ &P. \end{aligned}$$

Can we use resolution to establish R?

5. Using Logic to Verify an LLM-Generated Claim

Suppose an LLM is being used to answer questions about a small collection of known facts. The knowledge base contains:

$$\begin{aligned} &\text{Penguin}(P),\\ &\text{Penguin}(x) \rightarrow \text{Bird}(x),\\ &\text{Bird}(x) \rightarrow \text{Animal}(x). \end{aligned}$$

The two rules hold for every x. The LLM generates the statement:

> "Penguin P is an animal."

Can a logical verifier establish the claim?

Tutorial 4 – Probabilistic Intelligence

1. Latent variables (VAE)

For the same training image x, compare two VAE settings. The latent variable is one-dimensional with prior $p(z) = \mathcal{N}(0, 1)$.

Setting	Encoder distribution $q(z \mid x)$	Reconstruction loss R
A	$\mathcal{N}(1, 1)$	1.0
B	$\mathcal{N}(0, 1)$	1.8

Here $q(z \mid x) = \mathcal{N}(\mu, \sigma^2)$, with variance σ^2, and $R = -\mathbb{E}_{q(z|x)}[\log p(x \mid z)]$ is already calculated. Use natural logarithms and the formulas

$$D_{\mathrm{KL}}(q\|p) = \frac{1}{2}\left(\mu^2 + \sigma^2 - 1 - \log \sigma^2\right), \qquad \mathcal{L}_\beta = R + \beta D_{\mathrm{KL}}.$$

(a) Calculate the KL divergence for A and B.
(b) For $\beta = 1$ (the standard negative ELBO), calculate both losses. Which setting has the lower loss?
(c) Repeat for $\beta = 2$. Explain why the preferred setting changes even though the reconstruction losses stay fixed.

2. Latent variables (HMM)

Assume the following HMM. Hidden weather is hot (H) or cold (C). Observed ice-cream sales are high or low.

$$P(Z_1 = H) = P(Z_1 = C) = 0.5.$$

Transition probabilities

	$Z_{t+1} = H$	$Z_{t+1} = C$
$Z_t = H$	0.8	0.2
$Z_t = C$	0.3	0.7

Emission probabilities

	high sales	low sales
$Z_t = H$	0.8	0.2
$Z_t = C$	0.2	0.8

Each transition row gives $P(Z_{t+1} \mid Z_t)$. Each emission row gives $P(X_t \mid Z_t)$.

Observations. Day 1: high sales. Day 2: low sales.

(a) Identify the hidden variables and the observed variables for the two days.
(b) After observing high sales on Day 1, calculate

$$P(Z_1 = H \mid X_1 = \text{high}).$$

(c) Before using Day 2 sales, predict

$$P(Z_2 = H \mid X_1 = \text{high}).$$

(d) Now use the low sales on Day 2 to calculate

$$P(Z_2 = H \mid X_1 = \text{high}, X_2 = \text{low}).$$

Which weather state is more likely on Day 2?

3. Latent variables (Generative models)

Three systems learn from digit images without class labels:

System A Maps each image to one code and trains a decoder to reconstruct the same image.

System B Maps each image to a distribution over codes. Training balances reconstruction with closeness to a chosen prior.

System C Adds known noise to images and trains a network to predict that noise at different noise levels.

(a) Match A, B and C to a basic autoencoder, a VAE and a noise-prediction diffusion model.

(b) For B and C, describe how generation starts when no input image is available.

(c) A student samples standard Gaussian codes for A. Does good reconstruction guarantee good generated images? Why?

4. Adapting a response distribution

A pretrained LLM defines $P_\theta(y \mid x)$ for response y to prompt x. Fine-tuning uses prompts paired with concise technical-support responses. The loss uses response tokens only.

(a) For a training pair (x, y), explain what minimising

$$-\sum_t \log P_\theta(y_t \mid x, y_{<t})$$

encourages the model to do.

(b) After fine-tuning, we observe two problems. Name the issue in each case and explain the distinction.

- The model now performs worse on a previously successful general-writing test.
- Deployment prompts use unfamiliar terminology absent from the fine-tuning examples.

(c) Suppose LoRA freezes W_0 and learns $W = W_0 + BA$. Can the next-token distribution still change? Explain. Does this guarantee that the first problem disappears?